

Sanket patil / 37

Assignment 1

Explore following 4 websites and answer the questions

- bitcoin.org
- [Blockchain.com](https://blockchain.com/)
- <https://coinmarketcap.com/>
- <https://www.investopedia.com/>

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom?
Satoshi nakamoto
2. Bitcoin transactions are recorded on which data structure?
Blockchain
3. Bitcoin uses which consensus mechanism?
Proof of Work
4. The smallest unit of Bitcoin is called?
satoshi
5. Bitcoin is a **peer-to-peer** type of what system?
Electronic cash

Wallets

6. A wallet stores the user's **private** key key.
7. The recovery phrase used in wallets is called a **seed phrase** phrase.
8. Hardware wallets store keys **offline** (online/offline).
9. Bitcoin addresses are derived from **public** keys.
10. What is required to sign a transaction? **private** (public/private)

Security

11. Bitcoin prevents double-spending using **Blockchain**
12. Bitcoin is secured using **asymmetric** cryptography (asymmetric/symmetric).
13. To run a Bitcoin node, the software required is Bitcoin **core**
14. The process of verifying transactions is called **validation**
15. The Bitcoin network is described as **decentralized** (centralized/decentralized).

Mining

16. Miners solve **cryptographic** puzzles to add blocks.
17. Mining rewards a block with newly created **cryptocurrency coins**
18. The block time target for Bitcoin is **10** minutes.
19. Mining difficulty adjusts every **2016** blocks.
20. Miners compete to find a valid block **hash**

Blockchain Structure

21. Each block contains a **merkle** root.
22. Blocks are linked using the previous block's **cryptographic hash**

10. Write the TXID of the transaction. **8b7c5e30be488c97358cfc9447ca660f06d4fba941a1faba6db1d2c4adee0988**
11. Number of inputs:1
12. Number of outputs:1

13. Transaction fee: **0.00000330**

14. How many confirmations does it have? **960,835**

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block.

bc1qxy2kgdygjrqtzq2n0yrf2493p83kkfjhx0wlh

16. Record:

- Total BTC received: **16.56223521 BTC**
- Total BTC sent: **12.87005839**
- Current balance: \$230,941
What is the role of a full node?

List any two developer resources available on bitcoin.org.

Explore : <https://coinmarketcap.com/>

Task C1: Market Information

- Current price of BTC: 62,523.39
- 24-hour % change: 1.37
- Market capitalization: 1.25T
- Circulating supply: 20.06M BTC
- Total/maximum supply: 20.06M BTC
- **Task C2: Historical Data**
 - What was the price of Bitcoin one year ago? 113730
 - Note the highest and lowest prices in the last 30 days. 66700 / 58000

1. Why is Bitcoin considered volatile?

Bitcoin is considered **volatile** because its price changes rapidly in a short period due to factors such as market demand and supply, investor sentiment, news, government regulations, and macroeconomic events.

2. List two investment risks mentioned.

1. **High price volatility** – The value of Bitcoin can rise or fall significantly in a short time.
2. **Regulatory risk** – Changes in government laws or regulations can affect Bitcoin's price and usage.

Part E: Reflection & Analysis

1. Which website gave you the most clear understanding? Why?

The Bitcoin Whitepaper (Satoshi Nakamoto Institute) gave the clearest understanding because it explains how Bitcoin works directly from its creator. It clearly describes blockchain, peer-to-peer networking, cryptographic hashing, digital signatures, and Proof of Work without relying on third-party interpretations.

2. In one paragraph, explain how blockchain ensures trust without a central authority.

Blockchain ensures trust without a central authority by storing transactions in blocks that are cryptographically linked using hash values. Every participant in the network maintains a copy of the blockchain and independently verifies new transactions according to agreed consensus rules. Once a block is validated and added